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8 CAPABILITY DEVELOPMENT DOCUMENT
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12 DEPLOYABLE OXYGEN SYSTEM (DOS)
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Executive Summary

This Capability Development Document (CDD) updates the Operational Requirements Document, Deployable Oxygen System (DOS), AMC 001-00-B, 22 August 2001²³, to meet requirements identified in the Ground Contingency Medical Support System (GCMSS) Initial Capability Document (ICD), 16 October 2003²⁰, changes in Aeromedical Evacuation (AE) policy, and technological advances in oxygen generation and storage technology.

A capability is required to produce and store therapeutic oxygen for use in forward areas and in flight. Oxygen is a first-line therapeutic drug. The AE, ground-based, and special operations medical missions require medical support systems capable of providing therapeutic oxygen and operating oxygen-driven equipment in flight and on the ground. The Air Force (AF) has advanced the use of aircraft On-Board Oxygen Generation Systems (OBOGS), significantly decreasing the forward availability of Liquid Oxygen (LOX). AF deployed medical units are dependent on the aircraft cryogenic systems as a source of LOX. Finding reliable and safe oxygen in the battle space is increasingly problematic. In addition, personnel required for maintenance of liquid oxygen systems are not readily available in the theater of operation.

The DOS program delivers the capability to generate therapeutic oxygen in forward areas, significantly reducing supply and maintenance issues. Varying sizes of oxygen generators and storage systems allow for a tailored approach to providing oxygen. Large capacity generators will supply large field hospitals and fill portable storage vessels. Medium generators will deploy to smaller second echelon field medical facilities, while very small medical teams and Special Operations medical units will utilize small generators and storage vessels. In locations where oxygen generation is not feasible, such as in-flight AE missions, therapeutic oxygen storage vessels are used to deliver oxygen.

DOS addresses alternative material approaches by embracing the family of systems approach where therapeutic oxygen is generated and stored utilizing a variety of technologies. Major considerations for the analysis of material approaches included the reduction of airlift and increased capacity for far forward users. The multiple sizes of oxygen generators and storage vessels in the DOS program allow for tailored deployment of capabilities to match mission requirements, minimizing the logistical footprint and enhancing mobility.

Revision History

Change Number	Date of Change	Date Entered	Name of Person Making Change

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138 **Points of Contact**

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1. Capability Discussion.

1.1. Future Force Capability.

The joint medical mission includes provisions for air, sea, and ground-based medical support for the full spectrum of medical care in deployed scenarios, including wartime operations, deterrence and contingency operations, peacetime engagement, crisis response, humanitarian relief, and Special Operations. Both ground-based and Aeromedical (AE) medical support require therapeutic oxygen. The Deployable Oxygen System (DOS) program supports the Focused Logistics Joint Functional Concept by ensuring healthy and fit combat forces (Force Health Protection). Strategic National (SN) and Strategic Theater (ST) tasks this CDD supports include:

- SN 4.3.3 Coordinate Defense-Wide Health Services
- SN 6.6.4 Expand Health Service Support
- SN 8.1.5 Conduct Foreign Humanitarian Assistance and Humanitarian and Civic Assistance
- SN 8.2.3 Support Evacuation of Noncombatants from Theaters
- SN 9.2.1 Coordinate CBRNE Protection for Strategic Forces and Means
- ST 4.2.2 Coordinate Health Service Support
- ST 4.2.2.3 Manage Medical, Dental, and Veterinary Services and Laboratories and Supply

The DOS program directly supports the Ground Contingency Medical Support System (GCMSS) Initial Capability Document (ICD)²⁰ dated 16 October 2003 and AE of sick/injured patients, in peacetime and contingency operations, under the supervision of qualified AE and medical crewmembers via fixed wing aircraft, in accordance with Air Force Policy Directive 10-21.⁴

1.2. Overview of the Capability Gap.

Each Service is responsible for Manpower Equipment Force Packages (MEFPAK) in support of deployed operations. Oxygen is a first-line therapeutic drug. The AE, ground-based, and special operations medical missions must have medical support systems capable of providing therapeutic oxygen and operating oxygen-driven equipment in flight and on the ground. The current methods employed to meet these requirements include use of the Portable Therapeutic Liquid Oxygen System (PTLOX), Deployable Oxygen Generator System (DOGS), and/or high-pressure oxygen cylinders. Finding reliable and safe oxygen in the theater of operation is difficult and increasingly problematic. The PTLOX is becoming logistically unsupportable. The advancement in the aircraft On-board Oxygen Generation Systems (OBOGS) has significantly decreased the forward availability of Liquid Oxygen (LOX) resupply. Additionally, personnel required for maintenance of liquid oxygen systems are not readily available in the theater of operation. The Air Force Medical Service can no longer depend on line support for LOX and therefore must generate its own oxygen. The U.S. Army Medical Service does not use LOX nor does the U.S. Navy Medical Service except on its two hospital ships and aircraft carriers.

The DOS program delivers the ability to generate and store therapeutic oxygen in forward areas. Varying sizes of oxygen generators and storage devices allow for a tailored approach to providing oxygen. Large capacity generators will supply large field hospitals and fill portable storage vessels. Medium generators will deploy to smaller level II medical facilities in the field while very small medical teams and Special Operations medical units will utilize small generators and storage devices. In locations where generation is not feasible such as in flight on AE missions, therapeutic oxygen storage vessels are used to deliver oxygen. The current increment of the DOS program is targeted at specific solutions to generate, store, and deliver acceptable amounts of therapeutic oxygen to patients in one to three years. Increased capability to reduce size, weight, and power consumption are projected in four to eight years. Developing technologies are expected to evolve so that the required amount of oxygen will be delivered by increasingly smaller equipment.

The increasing capability of DOS better supports all areas of the operating environment to include immature and mature Joint Operating Areas (JOA). Examples of use in the deployed areas of responsibility include: level III medical facilities, smaller level II medical facilities, and Special Operations units as they intersperse within the battle space. Improved DOS products will also be utilized in all areas of the AE system (ground-based and in flight) in both peacetime and during contingencies.

2. Analysis Summary.

DOS addresses alternative material solutions utilizing the family of systems approach where therapeutic oxygen is generated and stored utilizing a variety of technologies. Major considerations for the analysis of material approaches were the reduction of airlift and increased capacity for far forward users. On 13 February 2003, a meeting to review and discuss the analysis of LOX sustainment, oxygen purity, and volume requirements took place, resulting in the MAJCOMs and AF/SG agreeing upon new standards for oxygen purity levels and volume of gaseous oxygen for an EMEDS.²⁶ Additionally, the liquid oxygen approach was rated least desirable due to logistics and cost issues.

The oxygen generators avoid transporting oxygen into and within the theater via traditional re-supply methods. Several studies of oxygen generators have highlighted successes and shortfalls in current production models, e.g., Portable Therapeutic Oxygen Concentration System Military Utility Assessment, September 2003²⁸, and Small Oxygen Concentrators, December 18, 2003.²⁹ Additionally, analysis of improved storage systems such as the Mobile Oxygen Storage Tank (MOST) has led to promising technologies and approaches to storage systems.²²

These analyses addressed reliability, availability, maintainability, and cost criteria of material alternatives. In addition to production and storage devices, the presence of improved oxygen delivery devices, e.g., masks and nasal cannulas, offers enhancement to, but not replacement of, these primary devices. Further evaluation of alternative technologies is being reviewed by the Human System Wing (HSW) System Program Office (SPO).

3. Summary Concept of Operations (CONOPS).

DOS supports Joint Focused Logistics concepts which include Joint Task Force Global Mobility CONOPS¹⁹ AFMS Concept Document² "Restoration of Health", AFMS Strategic Plan³³, and USAF Agile Combat Support initiatives for safe, effective therapeutics provided through a

reduced logistics footprint. DOS will afford extended oxygen capabilities, producing oxygen at remote locations while enhancing mission support requirements for augmented oxygen supply (refill stations). The system modularity will provide flexible options to the medical operators without severe weight and space liabilities for ground or air movement. Medical forces must be flexible and provide “state-of-the-art” routine and field combat casualty care in a wide range of operational environments. With a focus on sustaining warfighter mission performance, direct interface with sister service medical teams and assets is necessary to effectively function within established theater medical care systems.

The operational outcome provided by the DOS program is force health restoration. Through the provision of therapeutic oxygen, medical personnel provide essential care in theater and enhanced care during evacuation to definitive care. To achieve the desired operational outcome, the DOS program must be able to provide therapeutic oxygen in sufficient quantities across the entire range of military operations and at all levels in the area of responsibility. This is done through a combination of oxygen generation and storage.

Enabling capabilities include power production for oxygen generation and readily available maintenance support provided by deployed biomedical equipment technicians and military treatment facilities; their efforts will be augmented by contractor depot-level support in Continental United States (CONUS) if it is determined to be in the best financial interest of the government.

DOS complements the integrated joint war-fighting force. Therapeutic oxygen is a first-line drug used in treating and stabilizing injured and ill persons that is used by all Services to provide effective force health treatment. The DOS program is designed to provide therapeutic oxygen in a self-sustaining manner.

4. Threat Summary.

DOS is not likely to be a primary target; however, it is subject to collateral destruction and damage from conventional weapons and degradation by contamination caused by nuclear, biological, and chemical (NBC) effects. Collateral damage to the electrical power distribution system will have the same effect as direct collateral damage to the DOS.

DOS directly reduces readiness risks associated with remote or isolated operations as a result of enemy interdiction, operational locations, and other transportation constraints, e.g., weather. Internal threats to the safe production, storage, and delivery will be countered by engineering and operational controls.

5. Program Summary.

There is not one technical solution to currently meet all of the needs for medical units in the ground, air, and sea combined. Reduced availability and unreliability on non-medical sources of bulk liquid and gaseous oxygen has resulted in significant mission degradation. Non-availability is compounded by the fact that maintenance support plans are increasingly difficult and expensive to execute in a given area of responsibility. Requirements in the past have been partially met through a variety of solutions that were commercially available. This document outlines a ‘systems’ approach to identifying capabilities to solve oxygen deficiencies in the field.

DOS is capable of delivering medical grade oxygen in the theater of operation. Individual DOS components address specific requirements for various deployment scenarios and basing through the battle area to include far forward based operations. An oxygen storage system will provide a stated capability, while an oxygen generation system will refill the storage system and generate a constant level of oxygen in the theater of operations. DOS products are integral and interoperable with other components to produce a force enhancement effect for higher demands of oxygen. As a logistically independent delivery system of oxygen, it is self-sustaining with a marked decrease in non-medical maintenance resources. This ideally results in high levels of patient care and successful mission execution.

The DOS is driven by real world events and the ongoing reduction of institutional non-medical assets to support medical missions. Commercial off-the-shelf solutions should be utilized in addition to programmed developmental efforts for evolutionary technologies. Incremental solutions are part of the program design. Future solutions encompass more efficient oxygen delivery and storage using emerging technologies. Examples of improved attributes include: increased storage and output ratios; reduced power requirements; conserving oxygen used, e.g., rebreathers; and packaged generation and storage systems to meet medical missions throughout the theater.

6. System Capabilities Required for the Current Increment.

Threshold requirements are designated by [T] and the objective requirements are designated by [O].

6.1. Key Performance Parameters (KPPs).

6.1.1. Oxygen Quality: Must provide oxygen at 93% [T] and 99% [O] United States Pharmacopia (USP) concentration. The balance of delivered contents shall be argon or nitrogen, except for trace concentrations of other atmospheric gases, not exceeding percentage levels allowed in "Oxygen, USP." Rationale: The USP establishes officially recognized standards of quality for medical materiel and equipment which manufacturers must meet to sell their products in the United States. Oxygen for patient care must meet USP standards.

6.1.2. Food and Drug Administration (FDA) Approval: DOS systems will be approved by the FDA as a medical device in accordance with Federal law [T]=[O]. Rationale: All medical equipment/materiel must meet Federal regulatory requirements. Failure to meet this KPP means the system is not procurable.

6.2. Deployable Oxygen Storage/Delivery Attributes.

6.2.1. Capacity: The large oxygen storage vessel(s) will have an equivalent capacity of 28,000 gaseous liters [T] and 32,000 gaseous liters [O]. The medium oxygen storage vessel will have an equivalent capacity of 9,000 gaseous liters [T] and 15,000 gaseous liters [O]. The small oxygen storage vessel will have an equivalent capacity of 3,000 gaseous liters [T] and 5,000 gaseous liters [O]. Rationale: The equivalent capacity meets or exceeds the minimum oxygen requirement for medical elements under conditions of full utilization.

316 6.2.2. Delivery.

317 6.2.2.1. Large and Medium Storage Vessels: The storage vessels will be capable of delivering
 318 100 lpm at 50±5 pounds per square inch gauge (psig) [T] and 120 liters per minute (lpm) of
 319 gaseous oxygen at 50±5 psig [O]. Outlets: The storage vessel will provide a minimum of three
 320 oxygen outlets. Outlets will be compatible with oxygen hoses and flow meters currently in the
 321 Air Force Medical Service. Outlet Flow Rate: Each outlet will be capable of flow rates up to 15
 322 lpm and any one outlet will be capable of providing 100 lpm or greater. Rationale: Required
 323 oxygen delivery rates and pressures for this system will meet the requirements to drive existing
 324 gas-driven medical equipment. Flow rates are required to accomplish patient resuscitation
 325 procedures. Minimum flow rates of 10 lpm are further supported by American Heart
 326 Association cardiac life support guidelines, which require oxygen be delivered at flow rates
 327 greater than 10 lpm.

328 6.2.2.2. Small Storage Vessels: The small storage vessel will be capable of delivering 15 lpm
 329 at 50±5 psig [T] and 50 lpm of gaseous oxygen at 50±5 psig [O]. Rationale: Same as paragraph
 330 as 6.2.2.1.

331 6.2.3. Human Factors/Portability.

332 6.2.3.1. The large oxygen storage system will be man portable not to exceed 300 pounds (lbs)
 333 [T], 150 lbs [O] total weight when filled to capacity. These weights include storage/shipping
 334 containers. Use of retractable and lockable wheels may be used for ease of transport, movement,
 335 and maneuverability. Rationale: Required for operational use in larger second and third level
 336 deployable medical treatment facilities with minimal material handling equipment, e.g., forklifts.

337 6.2.3.2. The medium oxygen storage system shall be man portable not to exceed 150 lbs [T]
 338 and 37 lbs [O]. These weights include storage/shipping containers. Use of retractable and
 339 lockable wheels may be used for ease of transport, movement, and maneuverability. Rationale:
 340 For ease of movement in the deployed medical treatment facility.

341 6.2.3.3. The small oxygen storage system shall not exceed 35 lbs [T] and 17 lbs [O].
 342 Rationale: Paratroopers must be able to parachute the small storage system by wearing it or
 343 packing it in tandem for airborne operations (air droppable).

344 6.2.4. Size/Volume: Each medium and small storage system will fit within one litter stanchion
 345 position (21 inches by 7 feet by 2 feet) [T]=[O]. The size of the Large storage vessel will not
 346 exceed 36 cubic feet [T]=[O]. Rationale: The medium and small units will be utilized for AE
 347 missions aboard various aircraft and be in close proximity to the patient.

348 6.2.5. Fit into Ambulances and Ambuses: The storage system(s) will fit and operate in DoD
 349 ambulances, ambuses, and other "Personnel Recovery Vehicles (PRV)." In addition, the
 350 medium and small storage vessels will fit and operate in a PRV without impeding the operational
 351 medical efficiency [T]. All vessels will be capable of fitting into DoD side-loading ambuses or
 352 ambulances [O]. Rationale: Medical personal are required to transport and use storage devices
 353 in various vehicles.

6.2.6. Fit into Fixed and Rotary Wing Aircraft: The storage systems will fit through the various door openings on fixed- and rotary-wing aircraft and operate within the litter stanchion area of a fixed-wing AE aircraft without impeding egress, restricting routine aircrew operations, or taking up space in the aisles [T]=[O]. Rationale: Medical personal are required to transport and use storage devices in various aircraft.

6.2.7. Air Transportability: The storage system will be air transportable by fixed-wing cargo aircraft [T] and the CV-22 or rotary-winged aircraft [O]. Rationale: Medical personnel are required to transport these devices in various aircraft.

6.2.8. Altitude: The storage system will operate at an altitude pressure of up to 15,000 feet mean sea level (MSL) [T]=[O]. Rationale: DOS systems must be functional at locations with pressures less than sea level equivalent to meet operational demands. DOS systems will not be used at altitude pressures greater than 15,000 feet MSL.

6.2.9. Decompression: The storage system will operate safely during and after decompression from 8,000 feet to 45,000 feet altitude at 1, 7, and 60-second intervals [T]=[O]. Rationale: The storage vessels will be operational aboard aircraft and subject to rapid decompression.

6.3. Deployable Oxygen Generation System (DOGS) Attributes.

6.3.1. Pressure: The large and medium gaseous oxygen generators will deliver gaseous oxygen through a regulator that is capable of supplying between 50 to 100 psig [T]=[O]. The small gaseous oxygen generator will deliver gaseous oxygen at 50 ± 5 psig [O]. Rationale: This is the minimum delivery level required to support oxygen driven medical equipment.

6.3.2. DOGS Flow Generation.

6.3.2.1. Small DOGS Flow Generation: The gaseous oxygen generator will be capable of producing oxygen at a rate of 10 lpm [T] and 15 lpm [O]. Rationale: This level is required for individual use and achieving 100% inspired oxygen.

6.3.2.2. Medium DOGS Flow Generation/Compressor: The gaseous oxygen generator will be capable of producing oxygen at a rate of 60 lpm [T] and 180 lpm [O]. Rationale: These levels are necessary for oxygen requirements at level II or greater facilities.

6.3.2.3. Large DOGS Flow Generation/Compressor: The gaseous oxygen generator will be capable of producing oxygen at a rate of 180 lpm [T] and 300 lpm [O]. Rationale: These levels are necessary for oxygen requirements at level III or greater facilities.

6.3.3. DOGS Generator Weight.

6.3.3.1. Small DOGS Generator Weight: The weight of the small oxygen generator will not exceed 15lbs [T] and 1 lb/lpm of system weight [O]. Rationale: The generator must be light weight and able to fit in or on a medical rucksack or secured to a paratrooper.

6.3.3.2. Medium DOGS Weight: The weight of a 60 lpm oxygen generator will not exceed 800 lbs [T]. A medium DOGS which produces 61 lpm up to 180 lpm will use the following formula to determine the threshold weight $[(x)(13.333 \text{ lbs/lpm})](0.85) = \text{lbs}$, where $x = \text{lpm}$. For example, a system that produces 100 lpm will have a threshold weight of 1,133 lbs, $[(100 \text{ lpm})(13.333 \text{ lbs/lpm})](0.85)$ [T]. The weight of a 60 lpm oxygen generator will not exceed 500 lbs [O]. A medium DOGS which produces 61 lpm to 180 lpm will use the previous formula with a factor of 8.333 lbs/lpm to determine the threshold weight $[(x)(8.333 \text{ lbs/lpm})](0.85) = \text{lbs}$ [O]. Note: Weight includes generator and compressor. Rationale: These attributes represent an achievable goal in the reduction of weight while meeting the oxygen volume requirement for level II and III facilities. Additionally, if a system produces greater than 60 lpm, a 15% efficiency in weight to lpm output is required because of inherent system efficiencies.

6.3.3.3. Large DOGS Weight: The weight of a 180 lpm oxygen generator will not exceed 2,040 lbs [T]. A large DOGS which produces 181 lpm to 300 lpm will use the following formula to determine the threshold weight $[(x)(13.333 \text{ lbs/lpm})](0.85) = \text{lbs}$, where $x = \text{lpm}$. For example, a system that produces 250 lpm will have a threshold weight of 2833 lbs, $[(250 \text{ lpm})(13.333 \text{ lbs/lpm})](0.85)$ [T]. The weight of the 180 lpm oxygen generator will not exceed 1275 lbs [O]. A large DOGS which produces 181 lpm to 300 lpm will use the previous formula with a factor of 8.333 lbs/lpm to determine the threshold weight $[(x)(8.333 \text{ lbs/lpm})](0.85) = \text{lbs}$ [O]. Note: Weight includes generator and compressor. Rationale: These attributes represent an achievable goal in the reduction of weight while meeting the oxygen volume requirement for level II and III facilities. Additionally, if a system produces greater than 60 lpm, a 15% efficiency in weight to lpm output is required because of inherent system efficiencies.

6.3.4. DOGS Size.

6.3.4.1. Small DOGS Size: Must not exceed 1.6 cubic feet [T] and 0.026 cube feet [O]. Rationale: The generator must be able to fit in or on a medical rucksack or secured to a paratrooper's body.

6.3.4.2. Medium DOGS Size: Must not exceed 165 cubic feet at 120 lpm using the following formula $x + [(x-165)(-.333)] = \text{cubic feet}$ [T], where x equals lpm and 140 cubic feet at 120 lpm $x + [(x-140)(-.333)] = \text{cubic feet}$ [O]. Two medium units must fit on a standard 463L pallet. [T]=[O]. Rationale: The formula is based on a COTS oxygen generation system which is 165 cubic feet and delivers 120 lpm. The formula incorporates size efficiencies as a system produces greater than 120 lpm. Conversely, the formula allows for size inefficiencies as the lpm output reduced. Size is traded for output and weight improvements.

6.3.4.3. Large DOGS Size: Must fit within one half (1/2) the length, width and height dimensions of a United States Air Force (USAF) 463L pallet [T] and one quarter pallet [O]. Rationale: Limited lift available and palletization requirements.

6.3.5. The DOGS small version will be operable inside the EMEDS tent system [T]=[O]. Rationale: Needed to operate in a tent system for individual patient use.

6.3.6. Generator Fill Rate (Medium and Large System): The gaseous oxygen generator will have the capability to fill oxygen cylinders at a fill rate of 50% [T] and 100% [O] of the

428 maximum oxygen generator flow rate. This system will be capable of continued operations for
 429 filling and delivery simultaneously. Rationale: Level established to fill cylinders rapidly.

430 6.3.7. Power Characteristics: The DOS components requiring external electric power for
 431 operation will be compatible with the Ground-Based Deployable Power Generation and
 432 Distribution System (DPGDS). For compatibility with these power sources, the DOS will be
 433 designed for operation on 110-220 VAC, 50/60 Hz, single phase or 208-220 VAC, 50/60 Hz,
 434 three-phase power [T]=[O]. Rationale: Required to integrate with existing power sources.

435 6.3.8. Power Consumption: Small System – 1.0 kw [T] and 0.1 kw [O]; Medium System – 7.0
 436 kw [T] and 3.0 kw [O]; & Large System – 10.0 kw [T] and 5.0 kw [O]. Rationale: Small
 437 generator will be capable of operating with a self-contained power source, i.e., battery and 110
 438 VAC standard receptacle.

439 6.3.9. Power Connector: Small generator system must have a standard UL power connector
 440 [T]/[O]. Medium and Large generator system must have a power connector compatible with
 441 DoD 100 kw generator using a 60 amp Class-L electrical power connector [T]=[O]. Rationale:
 442 Required to integrate with existing power sources.

443 6.3.10. Alarms: All generators shall have audible and visual alarms for low oxygen purity and
 444 low oxygen pressure. Alarms will meet American Society for Testing and Materials (ASTM)
 445 standards such as F1463-93 (1999) Standard Specification for Alarm Signals in Medical
 446 Equipment Used in Anesthesia and Respiratory Care⁵ [T]=[O]. Rationale: Patient health and
 447 safety issue.

448 6.3.11. Outlets: A minimum of two outlets will be compatible with Compressed Gas
 449 Association (CGA) standards for oxygen hoses and flow meters [T]=[O]. The small generator
 450 will deliver oxygen via one outlet [T]=[O]. Rationale: Capability and logistics issue.

451 6.3.12. Accessories: The medium and large generators will have an accessory kit for each
 452 gaseous oxygen generator for delivery/distribution to a manifold distribution system. The kit
 453 will include standard medical grade oxygen hose(s) [100 feet minimum] and will be
 454 interoperable with DOS storage vessels. The kit will be secured to gaseous oxygen generator
 455 [T]=[O]. Rationale: Capability and logistics issue.

456 6.3.13. Emergency Operation: In the event the primary system fails to generate oxygen for any
 457 reason, the medium and large gaseous oxygen generator will automatically supply oxygen by a
 458 secondary system to patients for two (2) hours [T] and eight (8) hours [O] at normal system
 459 parameters. Rationale: Required operations in system failure conditions. Provide oxygen
 460 during main system repair.

461 **6.4. Common DOS Attributes.**

462 6.4.1. Accessories: The DOS system will have appropriate gauges, regulators, flow meters, and
 463 indicators for monitoring delivery and generation of oxygen [T]=[O]. Rationale: ASTM
 464 standard for oxygen systems.

465 6.4.2. Humidity: The DOS must be capable of operation in humidity levels up to 94±4% [T],
 466 100% [O]. Rationale: Field operational requirement.

467 6.4.3. Temperature Operation Range: The DOS must be capable of operating in temperatures
 468 ranging from -25F to 120F [T] and -65°F to 160°F [O]. Rationale: Realistic field operating
 469 temperature range.

470 6.4.4. Temperature Storage Range: The DOS must be capable of operations after being stored
 471 in temperature ranges from -40°F to 140°F [T] and -65°F to 160°F [O]. Rationale: Typical war
 472 reserve materiel storage range.

473 6.4.5. Air Transportability: The DOS system will be air transportable on fixed-wing [T] or
 474 rotary wing aircraft [O]. Rationale: Required to transport to theater and in the battle space.

475 6.4.6. Vibration /Shock: The storage vessel(s) will operate during and after being subjected to
 476 rapid decompression, vibration, shock, and acceleration as encountered during air, sea, and
 477 ground transportation. All other DOS components will operate after being subjected to
 478 vibration, shock, and acceleration as encountered during air, sea, and ground transportation
 479 [T]=[O]. Rationale: Ruggedization required for operation after shipment and during operation.

480 6.4.7. Power requirements (Aircraft): NA.

481 6.4.8. Securing Method: The DOS will be capable of being secured to floor provisions during
 482 flight operations on the following aircraft C-5, C-17, C-130, C-141, KC-10, KC-135, C-40 and
 483 CV-22. Additionally, it should be capable of being secured to C-2A, HH-46D, CH-46E, CH-47,
 484 CH/MH-60 series, HH-60 series, UH-60 series UH-1N, KC-130 series, C-9N and associated
 485 Army aircraft. DOS will be compatible with a standard USAF 463-L cargo pallet; D-Rings and
 486 5,000 lb cargo tie down straps [T]=[O]. Securing the storage vessel will be accomplished while
 487 wearing arctic, chem/bio, or flight gloves in a maximum time of ten (10) minutes [T] and five (5)
 488 minutes [O] for both fixed and rotary wing aircraft. Rationale: DOS systems will be transported
 489 on cargo aircraft and some systems will be operational aboard aircraft.

490 6.4.9. Storage Environment: DOS devices will be capable of withstanding long-term storage in
 491 a facility that is not environmentally controlled. The storage configuration and packaging will
 492 provide the capability for periodic inspection, preventive maintenance, and corrective
 493 maintenance. Oxygen Storage systems and Small DOGS will be capable of being removed from
 494 long-term storage, set-up, and fully operational within forty-five (45) minutes [T] and twenty
 495 minutes (20) [O]. The Medium and Large DOGSs will be capable of being removed from long-
 496 term storage, set-up, and fully operational within ninety (90) minutes [T] and sixty (60) minutes
 497 [O]. Rationale: System must be in war reserve storage for long periods of time, but will be
 498 required for use immediately.

499 6.4.10. Logistics and Readiness: The DOS system life will be an operational life of ten (10)
 500 years [T] and fifteen (15) years [O] and a storage life of fifteen (15) years [T]=[O]. System
 501 design will provide for ease and speed of maintenance and upgrade and minimize the logistics
 502 tail. System components will be easy to access and maintain. Removal, repair, replacement,
 503 adjustment, modification, checkout, and calibration will be accomplished using common shop

tools. Any special tools will be kept to a minimum and, as necessary, delivered as part of the support package. The contractor shall develop Support Equipment Recommendations Data (SERD) detailing the recommended test equipment and testing applications. SERDs shall not be submitted for the same item of equipment or for equipment currently in the Air Force inventory. Calibration and Measurements Requirements Summary (CMRS) shall be developed by the contractor only for items submitted on SERDs and designated by the government for CMRS submission. Specific maintenance career fields will be determined when the specific maintenance concept has been identified. Rationale: Sustainment of medical equipment is a responsibility of the Air Force Medical Service (AFMS) Logistics System.

6.4.11. Reliability/Maintainability: DOS Operational Availability (OA) will be 95% [T] and 98% [O], when operated continuously for a thirty (30) day period. Operational availability is necessary to ensure mission success. DOS mean time to repair (MTTR) for the large and medium gaseous generators will be 75% of the emergency oxygen operations attribute (para 6.3.19 Emergency Operation). The MTTR for the small system will be one hour. Rationale: The intent of the MTTR attribute is to allow organizational maintainers to repair the unit within the time allowed by the emergency operation backup system (under a majority of circumstances.) Gaseous generators will be equipped with an internal usage clock to track hours of operation. Time critical items, as identified by the manufacturer, will be specifically tracked. Time change items, as identified by the manufacturer, will also be tracked. Tracking clocks/counters will be able to record beyond the expected lifecycle of the item tracked. Storage vessel usage will be tracked based on calendar time. DOS will use standard off-the-shelf hardware, such as nuts, bolts, hoses, etc. [T]=[O]. Rationale: The inherent R&M characteristics of these components are expected to meet the needs of the user. The materiel developer will integration reliability/maintainability in to specific development milestones.

6.4.12. Electromagnetic Interference: DOS must operate in its intended electromagnetic environment without causing or suffering any unacceptable performance degradation due to electromagnetic interference or disrupt other equipment in the same environment [T]=[O]. Rationale: Must be able to operate on fixed and rotary winged aircraft without interfering with aircraft navigation systems.

6.4.13. Environmental Concerns: The DOS will be able to operate during and after being exposed to harsh environmental conditions such as blowing sand, snow, sleet, dust, rain, and high-salinity environments. The storage vessel must also operate within aircraft bank and angle of attack restrictions. Oxygen generators will operate on a slope of up to six (6) degrees. If any component of DOS is operated in a closed environment, venting will be at a level to prevent an oxygen enriched (hazardous) environment. The DOS will be capable of withstanding bacterial and/or viral disinfection with standard hospital disinfectants and bleach solutions without degradation of structural components or performance. The DOS will be decontaminable to the same degree as all other medical equipment items. When disposal of the DOS or its subsystems becomes necessary, the disposal process will not pose a threat to the environment. In accordance with DoDI 5000E3.T1¹¹, the program will incorporate environmental, safety, and health planning throughout the program life cycle [T]=[O]. Rationale: Ensures environmental quality concerns are addressed.

6.4.14. Conventional, Initial Nuclear Weapons Effects, Electromagnetic Environmental Effects (E3), and Nuclear, Biological, and Chemical Contamination Survivability (NBCCS), Conventional Effects: The DOS will be capable of performing under a wide range of potential battle-space environmental conditions and hazards (natural and man-made). DOS will be recoverable after the effects of chemical and biological contamination [T]. It is desirable that DOS will be operational during and after exposure to chemical and biological contamination [O]. Rationale: DOS supports the oxygen supply in the deployed treatment facility. Having a system that is quickly recoverable or operational through chemical or biological event is critical to the continuity of care.

6.4.15. Warranty: The DOS will be covered by an initial manufacturer's warranty for twelve (12) months [T] and thirty-six (36) months [O] that will include all parts and labor required for repair and maintenance. Rationale: Industry standard warranty meeting the service requirements.

6.4.16. Program Support: An integrated data package covering DOS storage, maintenance, inspection and integration will be delivered in digital format in accordance with DoD Standards. Data delivered digitally will be on CD ROM and in commercial desktop publishing format. The format will be stored, managed, and maintained in the Joint Computer Aided Acquisition and Logistics Support (JCALS) system. One (1) copy of DOS commercial technical manuals (operation/maintenance) will be provided per unit. Manuals will be formatted to enable update as required. [T]=[O]. Rationale: Minimum data support required from the manufacturer for an initial procurement.

6.4.17. Training Support: The training solution shall be delivered three months [T]/six months [O] prior to IOC. Rationale: Training required for effective/safe use and proper and timely repair.

6.4.18. Noise Levels: DOS systems will not exceed Air Force Occupational Safety and Health Standard (AFOSH) 161-20³ standards and MIL-STD-1472F¹⁵ [T=O]. Rationale: Health and safety.

6.5. DOS KPP Summary Table.

Paragraph	Key Performance Parameter	Development Threshold	Development Objective	Rationale and References
6.1.1.	Oxygen Quality	Oxygen at 93% USP concentration.	Oxygen at 99% USP concentration.	U.S Pharmacopia standard and AF/SG Memorandum ²¹
6.1.2.	FDA Approval	DOS articles will be approved by the FDA as a medical device.	Same	http://www.fda.gov/cdrh/ost/guidance/321.pdf ³²

Table 6.5. Key Performance Parameter Summary Table.

6.6. Deployable Oxygen Storage/Delivery Attribute Summary Table.

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.2.1.	Capacity	Large oxygen storage vessel(s) capacity of 28,000 gaseous liters. Medium oxygen storage vessel capacity of 9,000 gaseous liters. Small oxygen storage vessel will deliver 3,000 gaseous liters.	Large oxygen storage vessel(s) capacity of 32,000 gaseous liters. Medium oxygen storage vessel capacity of 15,000 gaseous liters. Small oxygen storage vessel will deliver 5,000 gaseous liters.	Compressed Medical Gases Guideline (Revised) February 1989 Prepared by: Center for Drug Evaluation and Research Food and Drug Administration³⁰

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.2.2.	Delivery	Large and medium oxygen storage vessel(s) capable of delivering 100 lpm at 50±5 psig. Small oxygen storage vessel capable of delivering 15 lpm at 50±5 psig.	Large and medium oxygen storage vessel(s) capable of delivering 120 lpm at 50±5 psig. Small oxygen storage vessel(s) capable of delivering 50±5 lpm at 50 psig.	Rationale: Required oxygen delivery rates and pressures for this system will meet the requirements to drive existing gas-driven medical equipment.
6.2.3.	Human Factors /Portability	Large oxygen storage system man portable not to exceed 300 lbs. Medium oxygen Storage system -man portable < 150 lbs. Small unit shall be < 35 lbs and wearable or packed in tandem for airborne operations.	Large storage system not to exceed 150 lbs. Medium unit shall be < 37 lbs. Small Unit shall not exceed 17 lbs and wearable or packed in tandem for airborne operations.	Required for operational use in larger second and third level deployable treatment facilities. For ease of movement on the deployed area. One man carry.
6.2.4.	Size/Volume	Medium and small storage unit will fit within the litter stanchion area. (21 inches by 7 feet by 2 feet). The large will be not more than 36 cubic feet.	Same	For on-board Aircraft use by AE requiring the oxygen storage systems in close proximity to the patient.

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.2.5.	Fit into Ambulances and Ambuses	All storage vessel(s) will fit and operate in DoD rear-loading ambulances and ambuses. The medium and small storage vessels will fit and operate in a Personnel Recovery Vehicle.	All vessels will fit into DoD side-loading vehicles.	Medical personnel are required to transport and use the storage devices in these various vehicles.
6.2.6.	Fit into Fixed and Rotary wing aircraft	All vessels will fit through the door openings on fixed and rotary-wing aircraft.	Same	Medical personnel are required to transport and use the storage devices in these various vehicles.
6.2.7.	Air Transportability	All Vessels will be Air transportable on fixed-wing cargo aircraft.	Rotary wing/CV-22	Medical personnel are required to transport these devices in various aircraft.
6.2.8.	Altitude	The storage system will operate at a cabin pressure equivalence of up to 15,000 feet MSL.	Same	DOS systems will be transported and used at cabin pressures equivalent to altitudes up to 15,000 feet MSL.
6.2.9.	Decompression	The storage system will operate safely during and after decompression from 8,000 feet to 45,000 feet cabin altitude at 1, 7, and 60-second intervals.	Same	The storage vessels will be operational aboard aircraft and subject to rapid decompression.

Table 6.2. Deployable Oxygen Storage/Delivery Attributes Summary Table.

582 **6.7. Deployable Oxygen Generation System (DOGS) Attributes Summary Table.**

583

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.3.1.	Pressure	Large and medium generators will produce gaseous oxygen through a regulator that is capable of supplying between 50 to 100 psig.	Small generator will produce gaseous oxygen at 50 ± 5 psig.	This is the minimum delivery levels required to support oxygen-driven medical equipment.
6.3.2.	DOGS Flow Generation	Small generator will deliver oxygen at 10 lpm. Medium generator will produce oxygen at a rate of 60 lpm. Large generator will produce oxygen at a rate of 180 lpm.	Small generator will deliver oxygen at 15 lpm. Medium generator will produce oxygen at a rate of 180 lpm. Large generator will produce oxygen at a rate of 300 lpm.	Level required to achieve 100% inspired oxygen. These levels are necessary for individual use, level II and III facilities, respectively.

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.3.3.	Generator Weight	Small oxygen generator weight will not exceed 15 lbs. Medium oxygen generator weight will not exceed 800 lbs @ 60 lpm otherwise $[(x)(13.333 \text{ lb/lpm})](0.85) = \text{lbs}$, where $x = \text{lpm}$. Large oxygen generator weight will not exceed 2,040 lbs @ 180 lpm otherwise $[(x)(13.333 \text{ lb/lpm})](0.85) = \text{lbs}$, where $x = \text{lpm}$.	Small oxygen generator weight will not exceed 1 lb/lpm. Medium oxygen generator weight will not exceed 500 lbs @ 60 lpm otherwise $[(x)(8.333 \text{ lb/lpm})](0.85) = \text{lbs}$. Large oxygen generator weight will not exceed 1,275 lbs @ 180 lpm otherwise $[(x)(8.333 \text{ lb/lpm})](0.85) = \text{lbs}$.	These attributes represent an achievable goal in the reduction of weight while meeting the oxygen volume requirement for level II and III facilities. Additionally, if a system produces greater than 60 lpm, a 15% efficiency in weight to lpm output is required because of inherent system efficiencies.
6.3.4.	DOGS Size	Small not to exceed 1.6 cubic feet. Must not exceed 165 cubic feet at 120 lpm using the following formula $x + [(x-165)(-.333)] = \text{cubic feet}$, where x equals lpm Large ½ dimension of a 463L pallet.	Small not to exceed 0.026 cubic feet. 140 cubic feet at 120 lpm $x + [(x-140)(-.333)] = \text{cubic feet}$ Large 1/4 dimension of a 463L pallet.	Based on a COTS oxygen generation system the formula incorporates size efficiencies as a system produces greater than 120 lpm. Conversely, the formula allows for size inefficiencies as the lpm output reduced. Size is traded for output and weight improvements.
6.3.5.	Operable Inside a Tent	The Small generator will be operable inside an EMEDS tent system.	Same	Ability to operate in a tent system enhances NBC and other environmental concerns.

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.3.6.	Generator Fill Rate	Large and medium generators will fill oxygen cylinders at a fill rate of 50% of the maximum oxygen flow rate.	Large and medium generators will fill oxygen cylinders at a fill rate of 100% of the maximum oxygen flow rate.	Level established to fill cylinders rapidly.
6.3.7.	Power Characteristics	Components requiring external electric power will be compatible with the DPGDS. DOS will operation on 110-220 VAC, 50-60 Hz, single phase or 208-220 VAC, 50-60 Hz, three-phase power.	Same	Required to integrate with existing power sources.
6.3.8.	Power Consumption	Small System 1.0 kw Medium System 7.0 kw Large System 10.0 kw	Small System 0.1 kw Medium System 3.0 kw Large System 5.0 kw	Small generator will be capability of operating with a self-contained power source.
6.3.9.	Power Connector	Small generator system must have a standard UL power connector. Medium and Large generators have connectors compatible with DoD 100 kw generators.	Same	Required to integrate with existing power sources.
6.3.10.	Alarms	Concentrators shall have audible and visual alarms for low oxygen purity and low oxygen pressure.	Same	Health and safety issue.

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.3.11.	Outlets	A minimum of two outlets will be compatible with CGA standards for oxygen hoses and flow meters. Small generators will deliver out of one outlet.	Same	Logistics and use issues.
6.3.12.	Accessories	Medium and large generators will have an accessory kit for delivery/distribution to a manifold distribution system. The kit includes standard medical grade oxygen hose(s) [100 feet minimum] and will be interoperable with DOS storage vessels. The kit will be secured to gaseous oxygen generator.	Same	Capability and logistics issue.
6.3.13.	Emergency Operation	Medium and large generators will automatically supply oxygen to patients for two (2) hours in the event of a primary system failure.	Medium and large generators will automatically supply oxygen to patients for eight (8) hours in the event of system failure.	Required operations in system failure conditions. Provide oxygen during main system repair.

Table 6.3. Deployable Oxygen Generation System (DOGS) Attributes Summary Table.

590 **6.8. Common DOS Attributes Summary Table.**

591

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.4.1.	Accessories	System will have appropriate gauges, regulators, flow meters, and indicators for monitoring delivery and generation of oxygen.	Same	ASTM standard for oxygen systems.
6.4.2.	Humidity	The DOS must be capable of operation and storage in humidity levels up to 94 $\pm$ 4%.	100%.	Field operational requirement.
6.4.3.	Temperature Operation Range	Capable of operating in temperatures ranging from -25° F to 120° F.	Capable of operating in temperatures ranging from -65° F to 160° F.	Realistic field operating temperature range.
6.4.4.	Temperature Storage Range	Capable of operations after being stored in temperature ranges from -40° F to 140° F.	Capable of operations after storage in temperature ranges of -65° F to 160° F.	Possible war reserve stock storage temperature range.
6.4.5.	Air Transportability	Air transportable on fixed-wing aircraft.	Rotary wing aircraft	Required to transport to theater and on the battle space.
6.4.6.	Vibration /Shock	Storage vessel(s) will operate during and after rapid decompression, vibration, shock, and acceleration. All other DOS components will operate after vibration, shock, and acceleration.	Same	Ruggedization required for operation after shipment and during operation.

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.4.7.	Power (Aircraft)	N/A	N/A	N/A
6.4.8.	Securing Method	Capable of being secured during flight operations on DoD cargo aircraft. Compatible with the AF 463-L cargo pallet, D-rings, and 5,000 lbs cargo tie down straps. Secure storage vessels while wearing arctic, chem/bio, or flight gloves in 10 minutes for fixed and rotary wing aircraft.	Secure storage vessels while wearing arctic, chem/bio, or flight gloves in a maximum time of 5 minutes for fixed and rotary wing aircraft.	DOS systems will be transported on cargo aircraft and some systems will be operational aboard aircraft.
6.4.9.	Storage Environment	Capable of operating after long-term storage in a facility that is not environmentally controlled. Oxygen Storage systems and DOGS Small capable of being removed from long-term storage, set-up, and fully operational within 45 minutes. Oxygen medium and large generators will be capable of being removed from long-term storage, set-up, and fully operational within 90 minutes.	Oxygen Storage system and DOGS Small capable of being removed from long-term storage, set-up, and fully operational within 20 minutes. Oxygen generators will be capable of being removed from long-term storage, set-up, and fully operational within 60 minutes.	System must be in war reserve storage for long periods of time, but will be required for use “immediately.”

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.4.10.	Logistics and Readiness	Operational life of 10 years and a storage life of fifteen 15 years. Basic maintenance using common shop tools.	Operational life of 15 years.	Sustainment of medical equipment is a responsibility of the AFMS Logistics System.
6.4.11.	Reliability/ Maintainability	OA of 95% over thirty days of continuous use. MTTR will be 75% of the emergency oxygen operation. Gaseous generators will be equipped with an internal usage clock to track hours of operation.	OA of 98% over thirty days of continuous use.	R&M characteristics of these components are expected to meet the needs of the user.
6.4.12.	Electromagnetic Interference	Operate in an electromagnetic environment without causing or suffering any unacceptable performance degradation.	Same	Must be able to operate on fixed and rotary winged aircraft.

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Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.4.13.	Environmental Concerns	Able to operate during and after being exposed to harsh environmental conditions. The storage vessel will operate within aircraft bank and angle of attack restrictions. If operated in a closed environment, venting will be at a level to prevent an oxygen enriched (hazardous) environment. Capable of withstanding bacterial and/or viral disinfection with standard hospital disinfectants. Decontaminable as a medical equipment item.	Same	Ensures environmental quality concerns are addressed.
6.4.14.	Conventional, Initial Nuclear Weapons Effects, E3 and NBCCS Effects.	Capable of performing under a wide range of potential battle-space environmental conditions and hazards (natural and manmade).	The DOS will be operational during and after exposure to chemical and biological contamination.	Quick recovery or operations through chemical or biological events is critical to the continuity of care.
6.4.15.	Warranty	Initial warranty for 12 months, including parts and labor required for repair and maintenance.	Initial warranty for 36 months, including parts and labor.	Industry standard warranty meeting the service requirements.

Paragraph	Attribute	Development Threshold	Development Objective	Rationale and References
6.4.16.	Program Support	Integrated data package covering DOS storage, maintenance, inspection, and integration.	Same	MIL-PRF-29612 Performance Specification; Training Data Product ²⁷ and MIL-HDBK-1379-1 DoD Handbook, Guidance for Acquisition of Training Data Products and Services. ¹⁴
6.4.17.	Training Support	Training delivered within three months prior to OIC.	Training delivered within six months prior to IOC.	Required for fielding and sustainment.
6.4.18.	Noise Levels	DOS systems will not exceed Air Force Occupational Safety and Health Standards	Same	AFOSH 161-20 ³ standards and MIL-STD-1472F15. ¹⁵

Table 6.4. Common DOS Attributes.

7. Family of System and System of System Synchronization.

The DOS fills the capability gap identified in the GCMSS ICD for delivery of oxygen but does not interface with any other Family of Systems or System of Systems. There are no other systems or programs supporting the DOS CDD. Anticipated changes in Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, and Facilities (DOTMLPF) are identified in paragraph 23. For the CDD crosswalk see Appendix A.

Capability	CDD Contribution	Related CDDs	Related CPDs
GCMSS ICD 4(c) 2	Supports restoration of health by providing oxygen for medical ground and AE platforms.	None	None

Table 7.1. Supported ICDs and Related CDDs/CPDs

8. National Security System & Information Technology System Supportability.

Not Applicable.

9. Intelligence Supportability.

Not applicable.

10. Electromagnetic Environmental Effects and Spectrum Supportability.

Not Applicable.

11. Assets Required Achieving Initial Operational Capability (IOC).

The number of DOS systems oxygen generators and storage vessels needed by the DoD medical community, including National Guard and Reserve units, is estimated in Table 11.1., which includes Navy and Air Force requirements. These numbers include initial spares and training sets. These numbers are IOC estimates. The exact number of DOS systems needed by the DoD medical community, including National Guard and Reserves units, can change based on the final manufactured solution(s).

	DOGS-Large	DOGS-Medium	DOGS-Small	Storage-Large	Storage-Medium	Storage-Small
Air Force	10	10	20	45	100	100
Navy	3	0	0	56	0	0
Total	13	10	20	101	100	100

Table 11.1. DoD Oxygen IOC Assets Required.

12. Schedule and Initial Operational Capability (IOC) /Full Operational Capability (FOC) Definitions.

IOC is scheduled for FY 2008. IOC will be achieved when 10 percent of the system is fielded to the initial units, unit personnel are trained, training base is established, and maintenance system is in place.

FOC is scheduled for FY 2013. FOC will be achieved when the system is totally procured and fielded, the identified force structure is trained, and a maintenance system is in place.

13. Other Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, and Facilities (DOTMLPF) Considerations.

The multiple sizes of oxygen generation and storage in the DOS program allow for tailored deployment of capabilities to match mission requirements, potentially minimizing the logistical footprint and enhancing mobility. The DOS program will not drive environmental quality compliance measures, nor safety and occupational health requirements. The DOS can be deployed to/within theater using normal materiel movement methods, i.e., 463L pallets. Most items are man portable. Large oxygen generators will require forklift support for placement. The only base operating support required is electricity for oxygen generators. No base operating support is required for storage/delivery systems.

- Doctrine: The DOS program will require some changes in Tactic, Techniques, and Procedures for deployed units. Overarching doctrine changes likely won't be required.
- Organization: The DOS program will not drive changes in the current organizational structure.
- Training: Limited training will be required for users of the equipment. This training can be accomplished at the unit level. Maintenance training will be required for the biomedical equipment repair career field. The total systems training concept and structure for the DOS must ensure delivery of timely and effective instructional methods, facilities, personnel, and tools for the life-cycle training at all levels and locations.
- Materiel: Depot repair will require interim contractor support until the Source of Repair Approval Process (SORAP) determines the depot repair source and that capability is established. The depot repair concept will be based on SORAP. It will be initiated by the Single Manager prior to Milestone B.
- Leadership and Education: The DOS program will not drive changes in leadership, but will require changes in education for use and repair.
- Personnel: Depending on the manufacturer's technical solution, the DOS program should not drive personnel changes. The DOS must be operated, maintained, and sustained by DoD and contractor personnel with the appropriate aptitudes, knowledge, skill levels, AFSC structures, anthropometrics, and force management factors.
- Facilities: The DOS program will not drive additional facility needs and does not require specialized supporting infrastructure or shelters.

654

655 **14. Other System Attributes.**

656 The DOS has several attributes that must be addressed in system design. These attributes include
 657 Human Systems Integration; Hazard of Electromagnetic Radiation to Ordnance; Wartime
 658 Reserve Materiel (WRM) requirements; natural environmental factors (such as climatic, terrain,
 659 and oceanographic factors); and unplanned stimuli (such as bullet impact).

660

Attribute	Development Threshold	Development Objective
Environmental Quality	Environmentally safe disposal. Leaks or outputs easily and safely cleaned.	Same
Human Systems Integration	Not adversely affect human systems. Human Systems Integration will be used to optimize the relationship between the human and the system.	Same
Wartime Reserve Materiel (WRM)	Will use standard WRM requirements. ¹²	Same
Accessibility	All parts must be easily accessible by operator and service personnel.	Same
Environmental Factors	Refer to MIL-STD-810F Explosive Atmosphere. ¹⁶	Same
Safety Hazards	Refer to ISO 8359 Oxygen Concentrators for Medical Use -- Safety Requirements. ²⁴	Same

661

Table 14.1. Other System Attributes.

662

663

664 **15. Program Affordability.**

665 Below is the current Deployable Oxygen System program funding profile. This profile is based
 666 on the current estimates of required quantities for oxygen storage vessels and oxygen generators.
 667 Pending the outcome of this CDD and developmental actions, the funding profile will be
 668 adjusted to reflect the new requirements.

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Current Program (\$M)	FY05	FY06	FY07	FY08	FY09	FY10	FY11
3600 RDT&E	3.4	2.9	2.2	2.2	2.2	0.0	0.0
3080 Procurement	0.0	0.0	1.2	3.1	2.1	2.1	2.1
3400 O&M	0.8	1.2	1.2	1.2	1.3	1.3	1.3

675

Table 15.1. Program Funding Profile.

676

677 **16. The Way Ahead.**

678

- AFROCC - Aug 04

679

- JROC following AFROCC, if required, prior to Milestone B

680

- Milestone B - Jan 05 (for initial system development)

681

- CPD prior to Milestone C

682

- Milestone C - Oct 2007

683

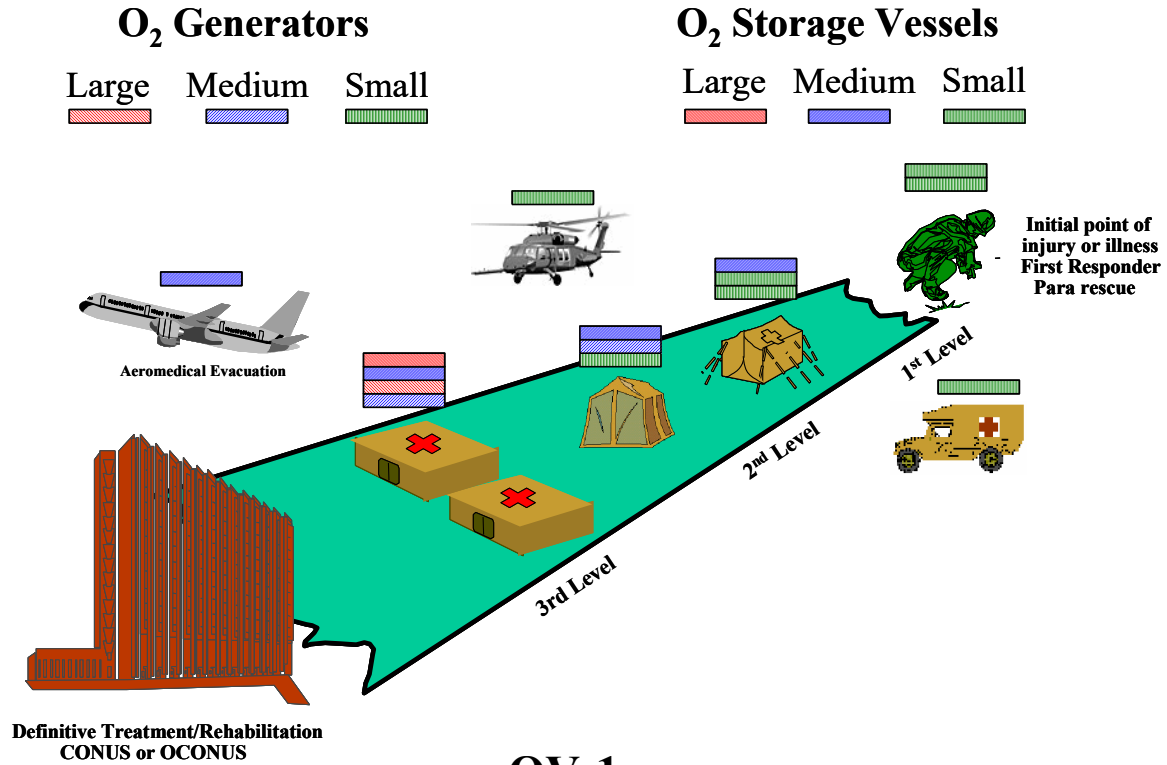
- Full rate production - Feb 2008

Appendix A CRD CDD/CPD/ORD/KPP Crosswalk Linkage

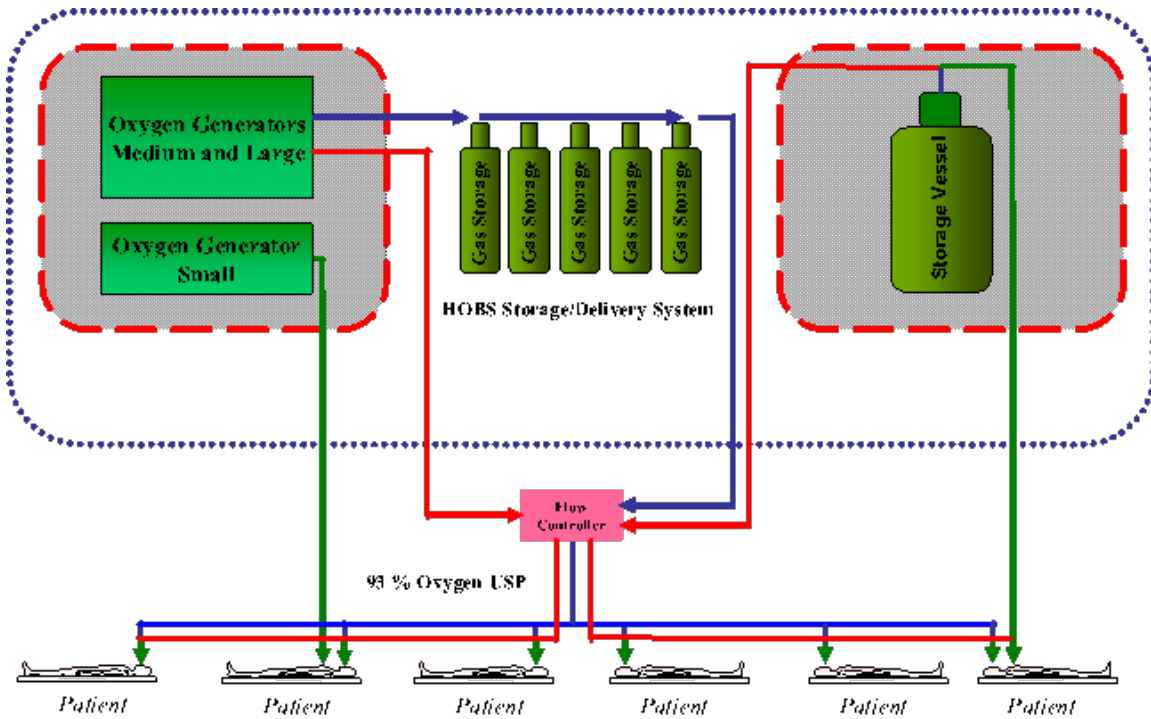
	TMIP CRD	GCMSS ICD
DOS CDD	There are no information technology/management issues associated with this CDD.	Supports restoration of health by providing gaseous oxygen for medical ground and AE platforms.

The Army recently brought forward an Operational Requirements Document (ORD) for a Medical Oxygen Concentrator (MOC).²⁴ This ORD represents a procurement of commercial off the shelf solutions to improve current capabilities. The Air Force is working closely with the Army in the source selection process.

Appendix B Architecture Framework Diagrams



OV-1



SV-1

Appendix C References

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Appendix D Acronym List

783		
784		
785		
786	ACC	Air Combat Command
787	AE	Aeromedical Evacuation
788	AF	Air Force
789	AFOSH	Air Force Occupational Safety and Health Standard
790	AFROCC	Air Force Requirements Oversight Capability Council
791	AOR	Area of Responsibility
792	ASTM	American Society for Testing and Materials
793	ATC	Air Transportable Clinic
794	CD ROM	Compact Disk Read Only Memory
795	CDD	Capabilities Development Document
796	CGA	Compressed Gas Association
797	CONOPS	Concept of Operations
798	COTS	Commercial Off-the-Shelf
799	CMRS	Calibration and Measurements Requirements Summary
800	CPD	Capabilities Production Document
801	CPEMEDS	Collectively Protected EMEDS
802	DoD	Department of Defense
803	DoDI	Department of Defense Instruction
804	DOGS	Deployable Oxygen Generation System
805	DOS	Deployable Oxygen System
806	DOTMLPF	Doctrine, Organization, Training, Materiel, Leadership & Education, Personnel &
807		Facilities
808	DPGDS	Deployable Power Generation and Distribution System
809	E3	Electromagnetic Environmental Effects
810	EMEDS	Expeditionary Medical Support
811	FDA	Food and Drug Administration
812	FOC	Full Operational Capability
813	GCMSS	Ground Contingency Medical Support System
814	HEMP	High-Altitude Electromagnetic Pulse

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815	HSW	Human System Wing
816	Hz	Hertz
817	ICD	Initial Capabilities Document
818	IOC	Initial Operational Capability
819	JCALs	Joint Computer Aided Acquisition and Logistics Support
820	JOA	Joint Operating Area
821	JROC	Joint Requirements Oversight Council
822	kW	Kilowatts
823	lbs	Pounds
824	lpm	Liters per minute
825	MEFPAK	Medical Equipment Force Package
826	MOC	Medical Oxygen Concentrator
827	MSL	Mean Sea Level
828	MTTR	Mean Time to Repair
829	NBC	Nuclear, Biological, and Chemical
830	NBCCS	Nuclear, Biological, and Chemical Contamination Survivability
831	OBOGS	On-Board Oxygen Generation Systems
832	ORD	Operational Requirements Document
833	O&M	Operation and Maintenance
834	PRV	Personal Recovery Vehicle
835	psig	Pounds Per Square Inch Gauge
836	PTLOX	Portable Therapeutic Liquid Oxygen System
837	RDT&E	Research, Development, Test, and Evaluation
838	SERD	Support Equipment Recommendations Data
839	SORAP	Source of Repair Assignment Process
840	SN	Strategic National
841	SPO	System Program Office
842	ST	Strategic Tactical
843	USAF	United States Air Force
844	USP	United States Pharmacopeia
845	VAC	Voltage Alternating Current
846	WRM	War Reserve Materiel

847 [O] Objective
848 [T] Threshold
849 °F Degrees Fahrenheit
850

Appendix E Glossary

Air Transportable Clinic (ATC): The ATC provides front line medical care for deployed forces. ATCs are capable of providing limited outpatient clinical care and emergency service for approximately 300-500 personnel for up to 30 days with minimal resupply. ATCs have the capability of providing basic stabilizing care for two trauma patients for a maximum of 12 hours and limited short duration field holding capability for non-trauma patients, not to exceed 48 hours. The ATC is not designed nor equipped to provide mass casualty treatment capabilities.

Level I: First Aid / Emergency Medical Care: Level I provides basic first aid: Self-aid and buddy aid, Combat Lifesaver, Combat Medic, and Basic Treatment squads.

Level II: Initial Resuscitative Care: Level II provides initial resuscitative care in the form of surgical and medical resuscitation. This care saves life and/or limb and stabilizes patients for evacuation to Level III. Blood and blood products are generally available at Level II. General surgeons and anesthesiologists or nurse anesthetists man the facilities. Other specialties may be represented. The ancillary support provided, particularly lab and radiology, is minimal.

At this level, care is rendered at the Army Clearing Station (division or corps), The Naval Surgical Company, and the Air Force EMEDS Basic and + 10. Here the casualty is examined and wounds and general status are evaluated to determine treatment and evacuation precedence. Emergency medical treatment, including beginning resuscitation, is continued and if necessary, additional emergency measures are instituted but they do not go beyond the measures dictated by the immediate necessities.

Level III: Resuscitative Care: Level III provides a higher level of surgical and medical resuscitative capability. In addition to general surgeons and orthopedists, other surgical specialists will be present. An example is the Combat Zone fleet hospital which has greater capabilities, particularly in laboratory and radiology support. The scope of treatment requires clinical capabilities normally found only in a hospital properly staffed, equipped, and located in an environment with a low level of threat from enemy action. This level of care constitutes the definitive treatment that is needed to return many patients to full duty.

Level IV: Definitive Care: In addition to the surgical capability provided in Level III, Level IV provides further definitive therapy for recovering patients who may then return to duty within the theater evacuation policy. A Communization Zone fleet hospital or an overseas medical treatment facility, i.e., Ramstein AFB, normally provides definitive care. This level of care is adapted to the precise condition of the patient; it is normally provided by a fully staffed hospital, delivering the care necessary to complete the patient's recovery.

Expeditionary Medical Support (EMEDS): The EMEDS system is a component of the Aerospace Medical Contingency Ground Support System. The Medical Contingency Ground Support System is composed of Unit Type Code (UTC) building blocks that provide personnel and equipment to meet specific operational requirements. These UTCs enable a deployed medical treatment facility to expand in modules to meet the full spectrum of theater requirements

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890 to include significant specialty care. These capabilities will be utilized to provide essential care,
891 deferring definitive care to CONUS or supporting theaters. Potential EMEDS deployments
892 include combat operations, deterrence and contingency operations, peacetime engagement, crisis
893 response, and humanitarian relief operations. The EMEDS system can expand to include the full
894 spectrum of Air Force medical capability in a theater of operations.